

Arithmetic and complexity

Strassen's asymptotic rank conjecture for tight tensors

Difficulty: extreme **Importance:** broadly interesting**Status:** Open

Rating rationale: Extreme because it predicts minimal asymptotic complexity for every concise tight tensor; broad importance follows from consequences for matrix multiplication and exponential-time combinatorial algorithms.

Topic: complexity of structured tensor powers

Context and notation

Over $\mathbb{C}$, the tensor rank $R(T)$ is the minimum number of pure tensors in an exact sum for T .

Problem statement

A tensor $T \in (\mathbb{C}^d)^{\otimes 3}$ is *concise* if each of its three matrix flattenings has rank d . It is *tight* if some choice of bases admits injective functions $a, b, c : \{1, \dots, d\} \rightarrow \mathbb{Z}$ with $a(i) + b(j) + c(k) = 0$ whenever the coefficient T_{ijk} is nonzero. Is

$$\lim_{m \rightarrow \infty} R(T^{\otimes m})^{1/m} = d$$

for every positive integer d and every concise tight T ? Tensor powers use corresponding-factor grouping; rank is over $\mathbb{C}$.

Why it matters

This is a structural conjecture about many bilinear computations, beyond the single matrix-multiplication family. [AC-04](#) is a named special tensor with independent literature treatment; this universal statement is not its equivalent reformulation.

References and status

A. Björklund and P. Kaski, *The Asymptotic Rank Conjecture and the Set Cover Conjecture are not Both True*, Conjecture 3 and §2.3, states the conjecture and gives a conditional consequence, not a refutation. K. Lee, *Asymptotic rank bounds: a numerical census* (2026), Conjecture 1, retains the tight/concise formulation. Searches for “asymptotic rank conjecture proved 2026” found no resolution. Numerical evidence is not a proof. **Admitted: no resolution located.**

Status check — 2026-09-10

Rechecked [Lee, Conjecture 1](#) and [Björklund–Kaski](#), then searched for 2026 proofs and counterexamples. The tight/concise asymptotic-rank formulation remains explicitly conjectural. Its incompatibility with the Set Cover Conjecture is conditional and does not refute it. No unconditional full resolution was located; numerical rank estimates do not prove the universal limit.

Reviewed research submission — 14 September 2026

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Theorems 3.2–9.3 give scoped power classifications and finite lower bounds, not the missing asymptotic upper bound. Inherited claims are excluded from this audit. **Open** is retained. AI assistance disclosed; no external human review, formal verification or Lean checks.